

MATHEMATICS FOR COMPUTING

Statistics

Descriptive Statistics, Frequentist Statistics
and Model Estimation

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Part I – Descriptive Statistics

- Descriptive statistics
- Histogram
- Sample mean
- Sample variance
- Measures of dispersion
- Order statistics
- Quartiles and percentiles
- Sample covariance
- Sample covariance matrix

Part II – Inference and Estimation

- Frequentist statistics
- Sampling
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- Confidence intervals
- Parametric model estimation
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- Parametric vs non-parametric

Meaning

- Descriptive statistics consists of methods used to organize, summarize and present data in a meaningful way.
- It focuses on describing the observations that are available.
- Numerical and graphical summaries make a dataset easier to understand.

Main idea: convert a collection of observations into a compact and interpretable summary.

| Main Methods of Descriptive Statistics

1. **Measures of central tendency** – describe a central or average value.
2. **Measures of dispersion** – describe how widely observations are spread.
3. **Graphical representation** – displays the distribution visually.
4. **Relationships between variables** – describes how variables vary together.

Examples from the topic:

- Mean and variance
- Histogram
- Covariance
- Order statistics

| Histogram – Definition

A histogram is a graphical representation of the distribution of numerical data.

- Data are divided into intervals called **bins**.
- The height of each bar represents the frequency of observations.
- Bars are adjacent to each other.
- It is used for numerical data.
- It helps us understand the shape and distribution of the data.

Unlike separated categorical bars, histogram bars represent continuous numerical intervals.

| Histogram – Example Data

Consider the marks obtained by 20 students:

42, 45, 48, 51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 71, 74, 76, 78, 82, 85, 91

The observations are grouped into class intervals.

Marks	Frequency
40–49	3
50–59	5
60–69	5
70–79	4
80–89	2
90–99	1

| Histogram – Reading the Example

From the frequency table:

- The interval 50–59 contains 5 students.
- The interval 60–69 also contains 5 students.
- The highest frequency is 5.
- Only one student scored between 90 and 99.
- Most students scored between 50 and 79.

Interpretation: the histogram provides a visual summary of how the students' marks are distributed.

| Sample Mean

The sample mean is the arithmetic average of all observations in a sample.
For a sample $X_1, X_2, \dots, X_n$:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- $\bar{X}$ = sample mean
- X_i = i -th observation
- n = number of observations

The sample mean gives the central or average value of the observed data.

| Sample Mean – Worked Example

Consider the marks of five students:

$$X = \{50, 60, 70, 80, 90\}$$

Here, $n = 5$.

$$\bar{X} = \frac{50 + 60 + 70 + 80 + 90}{5} = \frac{350}{5} = 70$$

Result: the average mark of the students is 70.

Sample Variance

Sample variance measures how much observations vary around the sample mean.
For a sample $X_1, X_2, \dots, X_n$:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- S^2 = sample variance
- X_i = individual observation
- $\bar{X}$ = sample mean
- n = sample size

Sample Variance – Calculation

For $X = \{50, 60, 70, 80, 90\}$, the mean is 70.

X_i	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
50	-20	400
60	-10	100
70	0	0
80	10	100
90	20	400
Total		1000

$$S^2 = \frac{1000}{5 - 1} = 250$$

| Mean and Variance – Interpretation

For the sample $\{50, 60, 70, 80, 90\}$:

$$\bar{X} = 70, \quad S^2 = 250$$

- The mean 70 represents the average mark.
- The variance 250 represents the spread of marks around the mean.
- A larger variance indicates greater dispersion.
- A smaller variance indicates that observations are closer to the mean.

| Measures of Dispersion

Measures of dispersion describe how widely observations are spread around a central value.

- **Range** – difference between maximum and minimum values.
- **Variance** – average squared deviation from the mean.
- **Standard deviation** – square root of variance.
- **Coefficient of variation** – measures relative variability.

Purpose: dispersion complements the mean by showing how much the observations vary.

| Order Statistics

Order statistics are the observations arranged in ascending or descending order. For a sample $X_1, X_2, \dots, X_n$, after arranging in ascending order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

- $X_{(i)}$ is called the i -th order statistic.
- $X_{(1)}$ is the minimum value.
- $X_{(n)}$ is the maximum value.
- The middle order statistic gives the median.

| Quartiles and Percentiles

Quartiles divide an ordered dataset into four equal parts.

- Q_1 = first quartile
- Q_2 = second quartile (median)
- Q_3 = third quartile

The interquartile range is:

$$IQR = Q_3 - Q_1$$

Percentiles divide data into 100 parts.

| Quartiles and Percentiles – Relationships

Important percentile relationships are:

Percentile	Corresponding quartile
25th percentile	Q_1
50th percentile	Q_2 (median)
75th percentile	Q_3

- Quartiles help describe the distribution of data.
- Percentiles indicate relative position within an ordered dataset.
- They are useful for detecting unusual observations.

| Sample Covariance

Sample covariance measures the direction of the linear relationship between two variables.

For variables X and Y :

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- X_i, Y_i are individual observations.
- $\bar{X}, \bar{Y}$ are the sample means.
- n is the number of observations.

| Sample Covariance – Example

Suppose study hours and marks are:

$$X = \{1, 2, 3, 4\}, \quad Y = \{40, 50, 60, 70\}$$

Their sample means are:

$$\bar{X} = 2.5, \quad \bar{Y} = 55$$

Using the covariance formula,

$$S_{XY} = \frac{1}{3} \sum (X_i - 2.5)(Y_i - 55)$$

The resulting covariance is:

$$S_{XY} = 18.33$$

Interpretation: the positive value indicates that the two variables increase together in this example.

| Sample Covariance Matrix

A sample covariance matrix summarizes the variances and covariances among several variables.

For two variables X and Y :

$$S = \begin{bmatrix} S_X^2 & S_{XY} \\ S_{YX} & S_Y^2 \end{bmatrix}$$

where:

- S_X^2 = variance of X
- S_Y^2 = variance of Y
- S_{XY} = covariance between X and Y

| Covariance Matrix – Structure

The covariance matrix has an important structure:

- The diagonal contains variances.
- The off-diagonal elements contain covariances.
- Covariance is symmetric, so $S_{XY} = S_{YX}$.

Therefore,

$$S = \begin{bmatrix} \text{variance of } X & \text{covariance} \\ \text{covariance} & \text{variance of } Y \end{bmatrix}$$

The matrix provides a compact representation of relationships among variables.

Frequentist Statistics

Frequentist statistics uses sample data to make statements about unknown population parameters.

- Parameters are treated as fixed but unknown.
- Samples are treated as random.
- Statistics are calculated from the observed sample.
- The sample mean can be used to estimate the population mean μ .

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Frequentist View – Parameters and Statistics

Population	Sample
Parameter describes the population	Statistic is calculated from the sample
μ is an example of a population mean	$\bar{X}$ is the sample mean
Parameter is fixed but unknown	Statistic changes from sample to sample

This distinction forms the basis for frequentist inference.

Sampling is the process of selecting a subset of observations from a population.

- **Population:** the complete group about which information is required.
- **Sample:** a smaller group selected from the population.

Example:

- A college has 5000 students.
- 200 students are selected for a survey.
- Population = 5000 students; Sample = 200 students.

| Sampling – Why a Sample is Used

The sample is used to draw conclusions about the entire population.

- Studying every member of a population may not be practical.
- A selected subset provides observations for statistical analysis.
- Sample statistics can be used to estimate population characteristics.

Population → Sampling → Sample → Statistical inference

| Sampling – Common Methods

1. **Simple random sampling:** every member has an equal chance of being selected.
2. **Systematic sampling:** every k -th member is selected.
3. **Stratified sampling:** the population is divided into groups and samples are selected from each group.
4. **Cluster sampling:** the population is divided into clusters and selected clusters are studied.

| Mean Square Error

Mean Square Error (MSE) measures the average squared difference between an estimator and the true parameter.

For an estimator $\hat{\theta}$:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

It can also be expressed as:

$$MSE = \text{Variance} + \text{Bias}^2$$

where

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

| MSE – Bias and Variance

The decomposition is:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + [Bias(\hat{\theta})]^2$$

- **Variance** measures variability of the estimator.
- **Bias** measures the difference between the estimator's expected value and the true parameter.
- MSE combines both sources of error.

A smaller MSE generally indicates better accuracy of an estimator.

MSE – Worked Example

Suppose the true parameter is:

$$\theta = 10$$

and an estimator gives 9, 10, 11.

Estimate	Squared error
9	$(9 - 10)^2 = 1$
10	$(10 - 10)^2 = 0$
11	$(11 - 10)^2 = 1$

$$MSE = \frac{1 + 0 + 1}{3} = \frac{2}{3} \approx 0.667$$

Interpretation: the estimates are, on average, reasonably close to the true value.

| Consistency

An estimator is called **consistent** if it approaches the true population parameter as the sample size increases.

Mathematically,

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

- $\hat{\theta}_n$ is the estimator based on a sample of size n .
- θ is the true population parameter.

| Consistency – Interpretation

Consistency means that increasing the sample size improves the estimator's reliability.

- With larger samples, the estimator tends to get closer to the true parameter.
- The sample mean $\bar{X}$ is a consistent estimator of the population mean μ under standard conditions.
- As sample size becomes larger:

$$\bar{X} \rightarrow \mu$$

Consistency is concerned with the behaviour of an estimator as n grows.

| Confidence Interval

A confidence interval gives a range of plausible values for an unknown population parameter.

For a population mean with known standard deviation:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

For a 95% confidence interval:

$$z_{\alpha/2} = 1.96$$

A confidence interval gives an estimate together with an indication of uncertainty caused by sampling.

| Confidence Interval – Components

The interval

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

contains three main components:

- $\bar{X}$ – sample estimate of the population mean.
- $\sigma/\sqrt{n}$ – standard error component.
- $z_{\alpha/2}$ – critical value corresponding to the confidence level.

The interval communicates both the estimate and sampling uncertainty.

| Confidence Interval – Example

Suppose:

$$\bar{X} = 50, \quad \sigma = 10, \quad n = 100$$

For a 95% confidence interval, $z_{\alpha/2} = 1.96$.

$$\begin{aligned} CI &= 50 \pm 1.96 \left(\frac{10}{\sqrt{100}} \right) \\ &= 50 \pm 1.96 \end{aligned}$$

Therefore,

$$CI = (48.04, 51.96)$$

| Parametric Model Estimation

Parametric estimation assumes that the population follows a known probability distribution described by a finite number of parameters.

- The distributional form is specified.
- Unknown parameters are estimated from sample data.
- The number of parameters is finite.

Example: if $X \sim N(\mu, \sigma^2)$, then μ and σ^2 are the unknown parameters that can be estimated.

| Parametric Models – Examples

Examples given in the topic include:

Distribution	Parameter(s)
Normal	μ, σ^2
Bernoulli	p
Poisson	λ

The chosen probability distribution provides a model for the population, while its parameters describe that model.

| Parametric Estimation – Example

Suppose the sample is:

10, 12, 14, 16, 18

For a normal model, the population mean μ is an unknown parameter. The sample mean is:

$$\hat{\mu} = \bar{X} = \frac{10 + 12 + 14 + 16 + 18}{5} = 14$$

Estimated parameter: $\hat{\mu} = 14$.

| Non-Parametric Model Estimation

Non-parametric estimation does not require assuming a specific probability distribution for the population.

- No specific distribution is imposed on the population.
- The approach is useful when the population distribution is unknown.
- It provides a flexible representation of the observed data.

Main idea: reduce dependence on distributional assumptions.

| Non-Parametric Methods

Examples include:

1. Histogram
2. Empirical distribution function
3. Kernel density estimation
4. Rank-based methods

These methods work with the observed data without requiring a specific parametric distribution for the population.

| Non-Parametric Estimation – Advantages

The source highlights three main advantages:

- Requires fewer distributional assumptions.
- Useful when the population distribution is unknown.
- More flexible for complex datasets.

The trade-off described here is between distributional assumptions and flexibility.

Parametric vs Non-Parametric Estimation

Feature	Parametric	Non-Parametric
Distribution	Specific distribution assumed	No specific distribution required
Parameters	Finite number of parameters	Flexible representation
Assumptions	More assumptions	Fewer assumptions
Flexibility	Less flexible	More flexible
Examples	Normal, Poisson	Histogram, KDE

| Descriptive Statistics vs Inferential Statistics

Descriptive Statistics

- Organizes and summarizes observed data.
- Uses numerical and graphical summaries.
- Examples: mean, variance and histogram.

Inferential Statistics

- Uses sample information to learn about a population.
- Involves estimation and uncertainty.
- Examples: confidence intervals and estimation.

Descriptive methods summarize; inferential methods generalize from samples.

| Properties of the Sample Mean

For observations $X_1, X_2, \dots, X_n$,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

- Every observation contributes to the mean.
- The mean is expressed in the same unit as the observations.
- It is useful for representing the central value of numerical data.
- The mean can be affected by unusually large or small observations.

The sample mean is a basic estimator of the population mean.

Variance: Why Square the Deviations?

Sample variance uses squared deviations from the sample mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

- Positive and negative deviations should not cancel each other.
- Squaring makes every contribution non-negative.
- Larger deviations receive greater weight.
- The denominator $n - 1$ is used for sample variance.

| Comparing Variability

Two samples may have the same mean but different spreads.

- A small variance indicates observations are relatively close to the mean.
- A large variance indicates greater dispersion around the mean.
- Standard deviation is the square root of variance.
- The standard deviation is measured in the original unit of the data.

$$s = \sqrt{S^2}$$

Covariance: Interpretation

For two variables X and Y ,

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}).$$

- Positive covariance: the variables tend to increase together.
- Negative covariance: one tends to increase as the other decreases.
- A value near zero indicates weak linear co-movement.
- Covariance describes direction, but its magnitude depends on the units of measurement.

Covariance Matrix: Structure

For two variables, the sample covariance matrix is

$$S = \begin{bmatrix} S_X^2 & S_{XY} \\ S_{YX} & S_Y^2 \end{bmatrix}.$$

- Diagonal entries are variances.
- Off-diagonal entries are covariances.
- Since covariance is symmetric, $S_{XY} = S_{YX}$.
- The matrix summarizes variability and pairwise linear relationships together.

| Sampling: Why Use a Sample?

A population may be too large or costly to study completely.

- A population is the complete group of interest.
- A sample is a smaller group selected from the population.
- Sample information is used to draw conclusions about the population.
- A good sampling method should provide useful information while reducing unnecessary effort.

Example: From 5000 college students, selecting 200 students gives a sample of size 200.

| Sampling Methods: Comparison

Method	Basic idea
Simple random	Every member has an equal chance of selection.
Systematic	Every k -th member is selected.
Stratified	Population is divided into groups and samples are taken from each group.
Cluster	Population is divided into clusters and selected clusters are studied.

| Confidence Interval: Factors Affecting Width

For a population mean with known standard deviation,

$$CI = \bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

The width depends on:

- Sample size n : increasing n reduces the standard error.
- Standard deviation σ : greater variability produces a wider interval.
- Confidence level: a larger confidence level requires a larger critical value and therefore a wider interval.

Confidence intervals communicate both an estimate and sampling uncertainty.

| Key Formulae – Descriptive Statistics

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\text{Range} = X_{\max} - X_{\min}$$

$$IQR = Q_3 - Q_1$$

$$CV = \frac{s}{\bar{X}} \times 100\%$$

| Key Formulae – Inference and Estimation

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

$$MSE = \text{Variance} + \text{Bias}^2$$

$$\hat{\theta}_n \xrightarrow{P} \theta \quad (n \rightarrow \infty)$$

$$CI = \bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

| Quick Revision – Descriptive Statistics

- Descriptive statistics organizes, summarizes and presents data.
- A histogram groups numerical observations into bins and shows frequencies.
- The sample mean gives the average value of observed data.
- Sample variance measures spread around the sample mean.
- Dispersion includes range, variance, standard deviation and coefficient of variation.
- Order statistics arrange observations by position.
- Quartiles and percentiles describe positions within ordered data.
- Covariance describes the direction of linear co-movement.
- A covariance matrix stores variances and covariances together.

| Quick Revision – Inference and Models

- Frequentist statistics treats parameters as fixed and samples as random.
- Sampling selects a subset from a population.
- MSE combines variance and squared bias.
- Consistency describes convergence of an estimator to the true parameter as sample size increases.
- Confidence intervals provide ranges of plausible parameter values with sampling uncertainty.
- Parametric estimation assumes a specified distribution with finite parameters.
- Non-parametric estimation does not require a specific population distribution.

| Conclusion

- Statistical methods provide a structured way to summarize and understand data.
- Measures such as mean, variance, quartiles and covariance describe different aspects of observed data.
- Sampling connects population-level questions with manageable datasets.
- MSE and consistency help describe estimator performance.
- Confidence intervals communicate estimates together with sampling uncertainty.
- Parametric and non-parametric estimation provide different modelling approaches.
- Statistical summaries, sampling and estimation work together to support data-based conclusions.

Statistics connects data, estimation and decision-making.

Thank You
